

Mahsa Vali

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SUMMARY

Final-year PhD candidate in Computer Science (defence Sep 2026) working at the intersection of **annotation-efficient** and **trustworthy** AI for medical image analysis. Develops active-learning, self-supervised, and uncertainty-aware methods spanning **foundation** and **vision-language models** for computational pathology and multimodal medical imaging (H&E, PET/CT, retinal). Skilled in building reproducible **PyTorch** pipelines and large-scale training on **HPC/SLURM** clusters.

EDUCATION

- **Ph.D. Candidate in Computer Science** 2022 – Present
Faculty of Mathematics and Natural Sciences, University of Cologne Cologne, Germany
- **M.Sc. in Electrical Engineering (Communication Systems)** Sep 2018 – Oct 2021
Faculty of Electrical and Computer Engineering, Isfahan University of Technology Isfahan, Iran
- **B.Sc. in Electrical Engineering** Sep 2013 – Sep 2017
Faculty of Electrical Engineering, Sheikhbahae University Isfahan, Iran

EXPERIENCE

- **University Hospital of Cologne (UKK), BozekLab** May 2023 – July 2026
Research Staff & Ph.D. Researcher Cologne, Germany
 - Developed **self-supervised representation learning** methods (**SimCLR**, contrastive learning) for stimulated Raman histology (SRH), enabling brain metastasis classification with limited annotations.
 - Developed a **slide-level uncertainty-guided active learning** framework for multi-class H&E histopathology segmentation, validated across **5 external cohorts**.
 - Built scalable, reproducible **PyTorch** training pipelines with **Distributed Data Parallel (DDP)** on **SLURM/HPC** clusters.
 - Assisted in the preparation and writing of research grant proposals submitted to external funding agencies.
- **University of Regensburg – Image Processing Lab** Oct 2025 – May 2026
Research Collaborator Regensburg, Germany
 - Analyzed hallucination, EOS prediction, and autoregressive decoding in **BioGPT** and **HistoGPT** for pathology report generation.
 - Developed decoding-level mitigation strategies.
- **University of Geneva** 2024
Visiting Researcher Geneva, Switzerland
 - Developed an uncertainty-guided **active learning** framework for **PET/CT** tumor segmentation using **Monte Carlo Dropout**, **SimCLR**, and **3D U-Net**.
 - Reduced annotation effort by **60%** while maintaining near fully supervised accuracy.
- **Durham University, School of Engineering** May 2021 – Apr 2022
Research Collaborator Newcastle, UK
 - Developed CNN-based segmentation and classification models with **Transfer Learning** for retinal image analysis using **Fundus** and **OCTA**.

SKILLS

- **Programming:** Python, PyTorch, NumPy, SciPy, Pandas, OpenCV, Scikit-learn, Git
- **Deep Learning:** CNNs, Vision Transformers (ViT), 3D U-Net, Foundation Models, Vision-Language Models, Self-Supervised Learning, Representation Learning, Active Learning, Transfer Learning, Contrastive Learning, SimCLR, Uncertainty Quantification
- **Medical Imaging:** Histopathology, Whole-Slide Imaging (WSI), PET/CT, MRI, DICOM, NiftI, MONAI (Familiar), Fundus Imaging, OCTA
- **Scientific Computing:** Linux, Docker, SLURM, HPC, Distributed Data Parallel (DDP), Jupyter
- **Research Areas:** Computer Vision, Medical Image Analysis, Multimodal AI, Data-Efficient Learning
- **Languages:** English (Professional), German (A2), Persian (Native)

- [J.1] Vali, M., Bozek, K., et al. (2026). **Semi-supervised Retinal Foundation Learning for Multi-class Classification of Optic Disc Neuropathies in Fundus Photographs**. *Translational Vision Science & Technology (TVST)*, *Per Review*.
- [J.2] Vali, M., Weng, Z., Moreau, N., Tolkach, Y., Bozek, K. (2026). **Slide-Level Active Learning Reduces Annotation Burden in H&E images**. *Per Review*.
- [J.3] Gao, Y., Vali, M., et al. (2024). **Combination of Deep and Statistical Features of Pathology Images to Classify and Diagnose the Degree of Malignancy of Prostate Cancer**. *Journal of Imaging Informatics in Medicine*, 1–19.
- [J.4] Papi, Z., Fathi, S., Dalvand, F., Vali, M., et al. (2023). **Auto-Segmentation and Classification of Glioma Tumors Using Deep Learning Based on MRI**. *Neuroinformatics*, 1–10.
- [J.5] Vali, M., Mohammadi, M., et al. (2023). **Differentiating Glaucomatous Optic Neuropathy Using Deep Learning Algorithms**. *American Journal of Ophthalmology*, 252, 1–8.
- [J.6] Vali, M., Nazari, B., et al. (2023). **CNV-Net: Segmentation and Classification of Choroidal Neovascularization Using OCTA**. *Diagnostics*, 13(7), 1309.
- [J.7] Abedi, I., Vali, M., et al. (2022). **A High-Resolution Chest CT-Scan Dataset for COVID-19 Diagnosis**. *BMC Research Notes*, 17(1), 32.
- [C.1] Vali, M., Bozek, K. (2026). **Structured Report Generation from Whole Slide Images via Organ-Conditioned Vision-Language Modeling**. In *Medical Image Computing and Computer Assisted Intervention (MICCAI) – Under Review*.
- [C.2] Vali, M., Arabi, H., Bozek, K. (2025). **Active Learning with Uncertainty-Guided Sample Selection for PET/CT Tumor Segmentation**. In *IEEE International Conference on Imaging Systems & Techniques (IST)*.
- [C.3] Vali, M., Jabbari, A., et al. (2022). **A Review of the Impact of Artificial Intelligence on Health Care Systems**. In *International Conference on Information Technology, Computer & Telecommunication*.
- [T.1] Vali, M. (Expected Sep 2026). **Machine Learning Approaches for Histopathology Data Analysis**. Doctoral Thesis, University of Cologne.

TEACHING & MENTORING

- **Graduate Student Supervision** 2023 – 2026
University of Cologne & International Collaborators
 - Co-supervised 2 M.Sc. thesis projects in medical image analysis, focusing on deep learning for image segmentation, classification, and model evaluation.
- **Workshop Instructor – Biomedical Image Processing** 2022
Student Association of Medical Physics, Isfahan University of Medical Sciences
 - Designed and delivered a hands-on workshop on biomedical image processing as the Scientific Representative of the Student Association.
- **Teaching Assistant** 2020 – 2021
Isfahan University of Medical Sciences
 - Assisted in teaching graduate-level courses in *Advanced Signal Processing (Wavelets)*, *Digital Signal Processing*, and *Advanced Biomedical Signal Processing*, including tutorials and assessment preparation.

INVITED TALKS & CONFERENCE PRESENTATIONS

- **Oral Presentation** – IEEE International Conference on Imaging Systems & Techniques (IST), France Oct 2025
- **Invited Speaker** – IRAVO Annual Meeting, Tehran, Iran Jan 2025
- **Oral Presentation** – CANTAR (CANCer TARget) Annual Retreat Meeting, Essen, Germany Sep 2023

REFERENCES

Dr. Hossein Arabi
 Senior Researcher
 University of Geneva, Switzerland
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 Research Collaborator

Dr. Soheyla Amirian
 Assistant Professor
 Pace University, USA
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 Former Advisor

Prof. Dr. Katarzyna Bozek
 Professor
 University of Cologne, Germany
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 Ph.D. Supervisor